

Chance-constrained water pumping managing power distribution network constraints

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Abstract—We formulate a chance-constrained optimization problem to schedule water distribution network (WDN) pumping subject to water and power distribution network (PDN) constraints while managing water demand uncertainty. In addition to an optimal pumping schedule, we also determine optimal control policy parameters used to compute real-time control actions to compensate for demand forecast error. The resulting problem includes nonconvex constraints, and so conventional solution approaches for chance-constrained problems do not work. We heuristically apply a scenario-based method and investigate the control policy's performance to ensure all WDN and PDN constraints are satisfied despite uncertainty. Through case studies with a detailed model of a coupled WDN/PDN, we find that WDN pumping can be scheduled and controlled to manage PDN voltage constraints and that the scenario-based method provides feasible real-time control actions for many realistic water demand scenarios but more work is needed to identify computationally tractable approaches with probabilistic guarantees.

Index Terms—chance-constrained optimization, demand response, distribution networks, water networks

I. INTRODUCTION

The power distribution network (PDN) and the water distribution network (WDN) are both critical infrastructures that are experiencing increasing levels of strain. The power network has increasing distributed, intermittent renewable energy sources and growing power demands. The water network is facing new challenges including rising water demands and more energy intensive methods of treating and pumping water due to shrinking fresh water supply [1]. These challenges complicate the operation of both networks and make it more challenging to ensure PDN and WDN constraint satisfaction.

Presently, the PDN and WDN are operated in a decoupled fashion despite their intrinsic spatial and temporal interdependence. WDN pumps are passive loads on the electricity network. Pumping water as needed at times of high demand can present a significant load to the electricity network and also be financially taxing to water utility companies. The operation of drinking and waste water networks makes up around 4% of the total world-wide electricity consumption, with pumping being a majority of the load [1]. Consequently, scheduling and controlling water pumping can help mitigate peak electric demands or help the PDN manage constraints (e.g., voltage limits) and balance supply and demand.

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The objective of this paper is to develop an approach to schedule water pumping subject to the constraints of the water and power distribution networks. Both networks are subject to uncertainty, for example, from stochastic water demand, power demand, and equipment outages. Here, we consider only uncertain water demand and we assume a control policy that allows the pumps to adjust in response to forecast error by modifying pumping operations. Our approach determines the nominal schedule of the pumps and the optimal control policy parameters. We assume pumps in the WDN are capable of quickly responding to control signals. The California Independent System Operator has established that industrial facilities, like water treatment plants, can provide fast demand response services [2]. Most WDNs are already equipped with communication and monitoring infrastructure that can be used to provide automated control signals [2].

A key challenge in optimizing power and water distribution networks is that these systems are nonlinear and the optimization problems are nonconvex. There is a significant body of work on strategies to minimize WDN pumping costs, e.g., [3]–[5]. We refer the reader to [6] for a review of optimal WDN control methodologies. There is also work on the optimal design or operation of a WDN under uncertainty, e.g., [7], [8]. Some of this work relies on simplified models and none of it considers the WDN's impact on PDN constraints. There is also a significant body of work on methods to solve the optimal power flow problem, see [9] for a review. Ref. [10] formulates and solves a chance-constrained optimal power flow problem for PDNs but does not consider the WDN.

There is growing interest in coordinating power and water network operation to reduce operational costs. In [11], [12], water pumping provides balancing services to the power network; however, the power network is not modeled and so the problems are essentially optimal water pumping problems. Ref. [13] formulates an optimal water pumping problem to minimize pumping costs and provide energy flexibility to the power transmission network (rather than the distribution network). Refs. [14], [15] develop a water-power flow problem and tractable solution algorithms while [16] proposes a water-power flow problem tailored to small-scale systems. However, these approaches do not consider uncertainty. The contributions of this paper are (1) formulation of the optimization problem to operate the WDN subject to the WDN and PDN

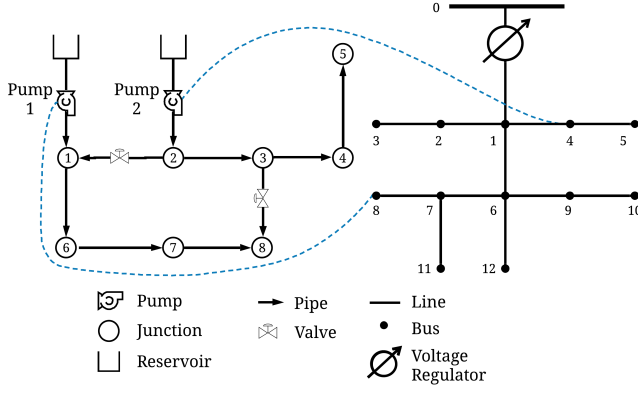


Fig. 1. Water (left) and power (right) distribution networks. Dashed lines show where water pumps are connected in the power distribution network.

constraints, (2) reformulation of the problem as a chance-constrained optimization problem considering water demand forecast error, (3) application of a scenario-based approach to solve the chance-constrained problem, (4) assessment of the impact of PDN constraints on optimal pumping, and (5) investigation of the approach's performance.

The remainder of the paper is organized as follows. Section II gives the problem description and assumptions. Section III formulates the deterministic and chance-constrained problems and describes the solution approach. Section IV describes our case study and its results. Finally, Section V discusses the limitations of our approach and directions for future work.

II. PROBLEM DESCRIPTION AND ASSUMPTIONS

In this paper, we develop an approach to schedule the pumps in a WDN to minimize pumping costs while ensuring WDN and PDN constraints are satisfied. Fig. 1 shows the topology of a coupled WDN and PDN. The pumps draw power from specific buses in the PDN. We first develop the deterministic optimization problem in which pumping costs are minimized subject to the water and power flow equations and inequality constraints limiting the hydraulic heads, pipe flows, voltages, and apparent power flows. We then reformulate the problem into a chance-constrained problem assuming uncertain water demand. We assume that the inequality constraints must hold probabilistically at a specified violation level, which is a design variable. In order to ensure that water demand is satisfied despite forecast error, we assume the pumps use an affine control policy to change their operation as a function of the total water demand forecast error. We optimize the control policy parameters together with the pump schedule. In this paper, we do not consider water storage tanks; however, they could be added to the formulation, increasing the number of constraints and decision variables since storage tank inflow/outflow can be scheduled and controlled in real-time like pumping.

The problem is solved over a single time period of length ΔT . The WDN is depicted as a directed graph $(\mathcal{N}, \mathcal{E})$ composed of sets of nodes $\mathcal{N}$ and edges $\mathcal{E}$. The nodes consist of a disjoint set of junctions $\mathcal{J}$ and reservoirs $\mathcal{R}$, i.e., $\mathcal{N} = \{\mathcal{J} \cup \mathcal{R}\}$. The edges that connect the nodes are pipes

that may have pumps ($\mathcal{P} \subset \mathcal{E}$) or valves ($\mathcal{V} \subset \mathcal{E}$); there is at most one pump or valve on a pipe. We denote a pipe $e : g \rightarrow h$ where the water flowing in pipe e goes from sending node g to receiving node h . We assume that the water flow in the pipes does not change direction within the time period and the flow is indicated by the directed graph. We also assume that the on/off states of pumps are determined in advance of the scheduling period, i.e., we do not consider the possibility of turning off the pumps.

We assume the PDN is radial and includes a set of buses $\mathcal{K}$ to which consumers and water pumps are connected. We also assume that the distribution network is balanced and pumping actions have no impact on electricity prices, i.e., the WDN is a price taker. Finally, we assume we have full knowledge of the WDN and the PDN, while in practice they are usually operated by different entities.

III. PROBLEM FORMULATION

In this section, we develop the problem formulation. We first formulate the deterministic problem. Then, we introduce uncertain water demand and formulate the chance constraints. Finally, we describe our solution approach using a scenario-based method.

A. Water Constraints

We use the the hydraulic constraints from [17]. We modify the formulation to include pressure reducing valves and reservoirs as defined in [15].

1) *Nodes, $\mathcal{N}$* : The total hydraulic head at node j , denoted H_j , is comprised of the elevation head $\bar{h}_j$ and the pressure head. It is constrained to ensure sufficient water pressure

$$h_j^{\min} + \bar{h}_j \leq H_j \leq h_j^{\max} + \bar{h}_j \quad \forall j \in \mathcal{N}. \quad (1)$$

Reservoirs are treated as an infinite source of water and, therefore, have a fixed pressure head. We set the reservoirs' hydraulic head equal to the elevation head

$$H_j = \bar{h}_j \quad \forall j \in \mathcal{R}. \quad (2)$$

For junctions, we denote the forecasted water consumption at junction j as d_j . We assume the water consumption within the time step is constant, non-negative, and there is no storage at the junction. The flow continuity constraint is

$$\sum_{e \in \mathcal{E}} a_{je} x_e = d_j \quad \forall j \in \mathcal{J}, \quad (3)$$

which ensures that the water consumed at junction j must equal the sum of the volumetric flow rate x_e in the pipes entering and exiting that junction. The parameter a_{je} is an element in the node-edge incidence matrix A which describes the connections in $(\mathcal{N}, \mathcal{E})$. Row j corresponds to node $j \in \mathcal{N}$ and column e corresponds to pipe $e \in \mathcal{E}$. Within this matrix, 1 or -1 at (j, e) designates that node j is the receiving or sending end of pipe e , respectively. The remaining entries are 0.

2) *Edges, $\mathcal{E}$* : The head loss along a pipe is equal to the head difference between the sending and receiving node

$$\sum_{j \in \mathcal{N}} a_{je} H_j = \begin{cases} \hat{H}_e - m_e(x_e)^2 & \forall e \in \mathcal{P}, \\ -L_e & \forall e \in \mathcal{V}, \\ -k_e(x_e)^{1.852} & \forall e \in \mathcal{E} \setminus (\mathcal{P} \cup \mathcal{V}), \end{cases} \quad (4)$$

where $\hat{H}_e$ and m_e are pump curve coefficients, and k_e is a resistance coefficient. The first case corresponds to pipes with pumps, the second to pipes with valves, and the third to the remaining pipes. For pipes with pressure reducing valves, the head loss $L_e \geq 0$ is a decision variable. The pipe flow is nonnegative and the pump flow is bounded

$$x_e^{\min} \leq x_e \leq x_e^{\max} \quad \forall e \in \mathcal{P}. \quad (5)$$

B. Power Constraints

The real and reactive power consumed at each bus are

$$P_{L,k} = \begin{cases} \rho_k + p_e, & \text{pump } e \text{ connected to bus } k \\ \rho_k, & \text{otherwise} \end{cases}, \quad (6)$$

$$Q_{L,k} = \begin{cases} \zeta_k + q_e, & \text{pump } e \text{ connected to bus } k \\ \zeta_k, & \text{otherwise} \end{cases}, \quad (7)$$

$\forall k \in \mathcal{K}$, where ρ_k and ζ_k are the real and reactive power demand of consumers connected to bus k , and p_e and q_e are the real and reactive power demand of pump e . While pumps are best modeled as motor loads, for simplicity, we assume that the pumps are constant power loads with a constant ratio of real to reactive power μ_e . To model the power flows, we use the branch flow model [18]

$$\sum_{n \in \mathcal{I}_{k+1}} P_n = P_k - r_k \frac{P_k^2 + Q_k^2}{V_k^2} - P_{L,k+1}, \quad (8)$$

$$\sum_{n \in \mathcal{I}_{k+1}} Q_n = Q_k - x_k \frac{P_k^2 + Q_k^2}{V_k^2} - Q_{L,k+1}, \quad (9)$$

$$V_{k+1}^2 = V_k^2 - 2(r_k P_k + x_k Q_k) + (r_k^2 + x_k^2) \frac{P_k^2 + Q_k^2}{V_k^2}, \quad (10)$$

$\forall k \in \mathcal{K}$, where P_k and Q_k are the real and reactive power flow downstream of bus k , r_k and x_k are the resistance and reactance of the line downstream of bus k , and V_k is the voltage at bus k . The set $\mathcal{I}_{k+1}$ consists of all lines downstream of and directly connected to bus $k+1$. The voltage at the feeder head is regulated to V_{set} and the remaining voltages are constrained

$$V_k^{\min} \leq V_k \leq V_k^{\max} \quad \forall k \in \mathcal{K}. \quad (11)$$

We could also bound the apparent power flows but in practice PDNs usually hit their voltage limits first.

C. Objective

Our objective is to minimize the scheduled energy costs of pumping water over all pumps in the WDN, i.e.,

$$\min_{\mathbf{x}} \sum_{e \in \mathcal{P}} \pi_e p_e \Delta T, \quad (12)$$

where π_e is the price of energy for pump e . The decision variable is $\mathbf{x} = \{x_e, H_j, L_e, P_k, Q_k, V_k\}$. The scheduled power consumption is a function of the flow rate x_e and head gain in the pump, which can be approximated by a quadratic function of x_e . For a fixed speed pump

$$p_e = \beta (\hat{H}_e - m_e(x_e)^2) x_e \quad \forall e \in \mathcal{P}, \quad (13)$$

where β is a constant. This equation is modified from the pump power consumption equation in [14], which uses variable speed pumps. The objective function is a cubic function of the flow.

D. Chance Constraints

We next consider uncertainty in water demand. We define $\tilde{d}_j$ as the water demand forecast error at junction j . We minimize the expected energy cost of pumping and since we assume the forecasts are unbiased, the objective function is equivalent to that in the deterministic case. The decision variable is expanded to $\mathbf{x} = \{c_e, x_e, H_j, L_e, P_k, Q_k, V_k\}$, where c_e is the control policy parameter of pump e . We assume an affine control policy that compensates for water demand forecast error by varying the output of supply pumps (i.e., pumps drawing from reservoirs not in-line booster pumps that increase pressure head)

$$\tilde{x}_e = c_e \sum_{j \in \mathcal{J}} \tilde{d}_j \quad \forall e \in \mathcal{P}, \quad (14)$$

$$\sum_{e \in \mathcal{P}} c_e = 1, \quad (15)$$

where $\tilde{x}_e$ is the real-time deviation of water flows in pipes with pumps from the scheduled flow. In (14), the total water demand deviation is multiplied by the control policy parameters to determine how the pumps should adjust in order to compensate the forecast error. In (15), the control policy parameters sum to one so that the changes in pumping operation fully compensate the water demand forecast error.

Using tilde ($\tilde{\cdot}$) to refer to deviations in variables resulting from the real-time control of pumps compensating water demand forecast error, we can write the stochastic counterparts of the deterministic constraints. For example, (1) becomes

$$h_j^{\min} + \tilde{h}_j \leq H_j + \tilde{H}_j \leq h_j^{\max} + \tilde{h}_j \quad \forall j \in \mathcal{N}.$$

The stochastic inequality constraints can be written as $f(\mathbf{x}, \tilde{\mathbf{d}}) \leq 0$ and transformed into a chance constraint

$$\mathbb{P}(f(\mathbf{x}, \tilde{\mathbf{d}}) \leq 0) \geq 1 - \epsilon \quad (16)$$

where the constraints must hold jointly with a probability of $1 - \epsilon$, where ϵ is the user-selected violation level. The resulting problem cannot be directly solved. Possible solution approaches include i) assuming specific uncertainty distributions, allowing the chance constraints to hold individually rather than jointly, and reformulating into an equivalent deterministic problem; ii) distributionally robust optimization; and iii) scenario-based approaches. However, since our problem includes nonlinear and nonconvex constraints, existing methods for i) and ii) do not work. Therefore, we resort to iii).

TABLE I
PUMP PARAMETERS

Pump	$\tilde{H}_e$ (m)	m_e (h ² m ⁻⁵)	$x_e^{\min}$ (CMH)	$x_e^{\max}$ (CMH)
1	100.5	0.0004	25	325
2	100.0	0.0001	50	700

E. Scenario Approach

We use the scenario approach [19] to solve our chance-constrained optimization problem. For convex problems, [19] gives a requirement for the number of randomly-selected scenarios N for which the stochastic constraints must hold to ensure feasibility at the chosen violation level [19]

$$N \geq \frac{2}{\epsilon} \left(\ln \frac{1}{\psi} + \delta \right), \quad (17)$$

where δ is the number of optimization variables and ψ is a user-selected confidence level. For example, for the case study presented in Section IV, a violation level of $\epsilon = 0.05$ and a confidence level of $\psi = 10^{-5}$ requires $N = 4,621$ scenarios. If the violation level is reduced to $\epsilon = 0.02$, the number of instances needed is $N = 11,552$. Since our problem is nonconvex, we apply the scenario approach heuristically; the guarantees derived in [19] no longer apply. Moreover, we test the approach with a varying number of scenarios with the goal of gaining insight into the performance and drawbacks of the method. In the future, we plan to explore the use of scenario-based approaches tailored to nonconvex problems, e.g., [20].

IV. CASE STUDY

A. Set-up

We use a modified version of the WDN presented in [4]; the topology is shown in Fig. 1. Ref. [14] also uses this network but makes different modifications than we do. Specifically, we model the pumps in the pumping station as a single pump with equivalent head and flow, we do not model the booster pump, and we model pressure reducing valves (using the model from [14]) instead of control valves. The WDN consists of 2 reservoirs, 8 junctions, and 11 pipes with 2 fixed speed pumps and 2 pressure reducing valves. The consumer water demand and $k_e \forall e \in \mathcal{E} \setminus (\mathcal{P} \cup \mathcal{V})$ are from [4]. The nodal elevation, minimum pressure head, and pressure reducing valve placements are from [14]. Since we do not model the booster pump nor add a surface tank, we changed the elevations at junctions 4, 5, and 6 (to 0, 20, and 40 m, respectively) and the minimum pressure head at junction 6 (to 10 m). Pump parameters are shown in Table I. We set $\beta = 6.97 \times 10^{-6} \frac{\text{kg h MW}}{\text{W m}^2 \text{ s}^3}$. The forecasted total water demand is 420 cubic meters per hour (CMH).

For the PDN, we use the topology of the IEEE 13-node feeder, shown in Fig. 1 [21]. Pumps 1 and 2 are connected to buses 8 and 4, respectively. The load at each bus is set to the average over the 3 phases. The distributed load along the line from bus 1 to 6 is placed at bus 1. The line impedance is calculated from the line lengths and the impedance of the first phase listed in the feeder specification sheet. The switch and transformer are each replaced with a line that has the

same impedance as the other segment on the same lateral. The shunt admittance and shunt capacitors are neglected. We explore three consumer power demand levels: 105%, 108%, and 110% of the nominal load at all buses. We set the price of electricity to $\pi_e = \$100/\text{MWh} \forall e \in \mathcal{P}$, and the line to neutral voltage at the feeder head to $V_{\text{set}} = 4.16$ kV. The voltage limits are $V_k^{\min} = 0.95$ pu and $V_k^{\max} = 1.05$ pu $\forall k \in \mathcal{K}$. The ratio of real to reactive power consumed by the pumps is $\mu_e = 3 \forall e \in \mathcal{P}$.

We use a Gaussian probability density function to randomly generate water demand forecast error with mean 0 and standard deviation $0.10d_j \forall j \in \mathcal{J}$. The distribution is truncated and scaled using an acceptance/rejection approach [22] such that only points within three standard deviations of the mean are included. We assume the errors at each junction are independent. We explore cases with $N = 10, 50, 100$, and 1000 scenarios. For cases $N = 10, 50$ we solved each case 100 times and for cases $N = 100, 1000$ we solved each case 10 times, and report the average results. Cases with fewer scenarios exhibit higher variability in results, requiring a larger number of solution instances to obtain reliable averages. The solver was unable to find a feasible solution in one instance for the 110% consumer power demand level and $N = 1000$ case. We generated another instance so that we could compute the average across 10 instances, but recognize that, if this occurred in a real system with real uncertainty scenarios, this could be problematic.

We solve the problem using the optimization package JuMP in Julia with Ipopt [23]. Since the problem is nonconvex and the solver can only guarantee a local minimum, we solve each problem 40 times using different initial guesses for the decision variables and chose the solution with the lowest cost.

B. Impact of Power Network Parameters On Pumping

We first analyzed the impact of PDN line parameters on the deterministic solution. The objective function is a concave function of the pumps' flow rates. The top left plot of Fig. 2 shows total pumping costs as a function of pump 1's flow rate. When the voltage constraints are not active, pump 1's optimal flow rate is at point A. As the line impedances increase, the voltages drop, approaching their lower limits. In the right two plots, we vary the line resistances r_k and reactances x_k as a percentage of their nominal values. The total pumping costs are shown in the top plot and the minimum bus voltages are shown in the bottom plot. Bus 8 always has the lowest voltage in the network. As the line resistance increases, the voltage at bus 8 gets closer to its lower limit of 0.95 pu. Once the voltage constraint becomes active, pump 1 must decrease its power consumption/flow rate to maintain the minimum bus voltage at 0.95 pu (i.e., move from point A toward point B), which increases pumping costs. This transition is shown by the light blue shaded area connecting the three plots. When pump 1 reaches point B, there is a second flow rate, i.e., point C, that yields the same total pumping cost. The minimum bus voltage increases when pump 1 switches to point C. The optimal flow rate for pump 1 remains at point C as the impedance continues

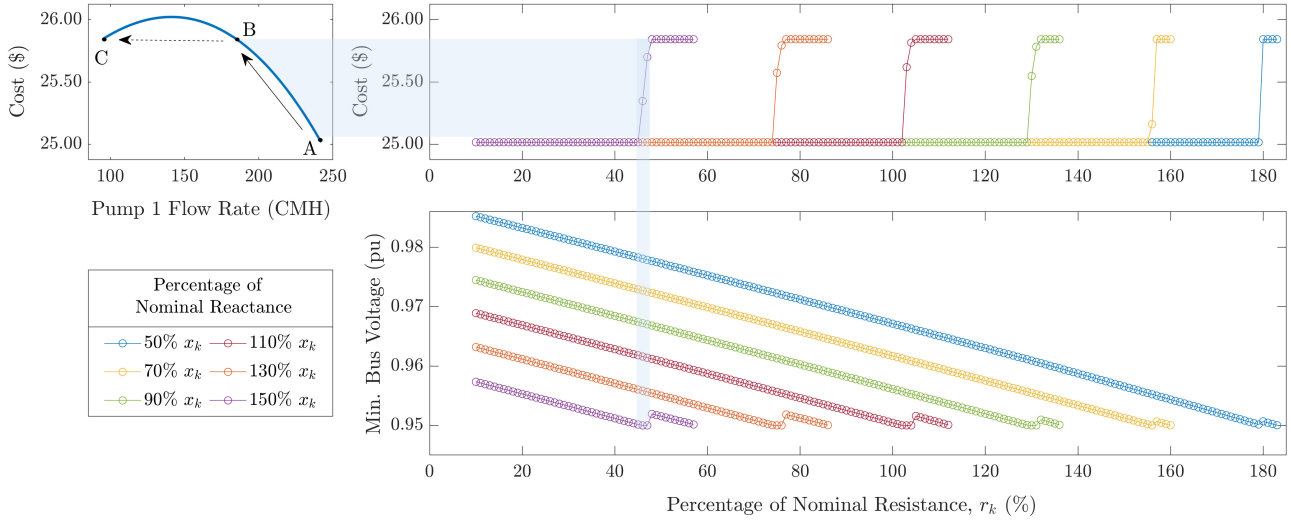


Fig. 2. Impact of distribution line parameters on pumping costs. (Top left) Pumping cost as a function of pump 1's flow rate and (Top right) as a percentage of the nominal line resistance and reactance. (Bottom right) Minimum bus voltage as a function of the percentage of the nominal line resistance and reactance.

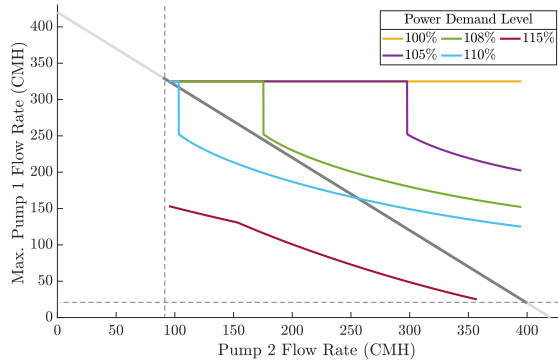


Fig. 3. Pump 1's maximum flow rate as a function of pump 2's flow rate for different consumer power demand levels.

to increase until the problem becomes infeasible. Note that when the PDN constraints become active and pumping is shifted from point A to point C, there is a 60% decrease in pump 1's flow rate. Pump 1 reduces its power consumption by 67.44 kW (51.23% power reduction).

All of the reactance curves in Fig. 2 exhibit the same behavior. Pump 1 moves from point A to point B when voltage constraints are active and resistances are increased, and then to point C when there is an equivalent cost. For higher reactance values, the solution will switch to point C for smaller resistance values (effectively shifting the curve left).

The effect of the consumer power demand level on the pump flow rates is shown in Fig. 3. The curves show pump 1's maximum flow rate as a function of pump 2's flow rate for different consumer power demand levels. Pump 1's maximum flow rate is constrained by PDN voltage constraints. The gray line shows all possible combinations of flow rates that meet the total forecasted water demand. The line is darkened for flow rates that satisfy (5). For the deterministic problem, the scheduled flow rates must fall on this darkened line. When

Power Demand (%)	$N = 10$	$N = 50$	$N = 100$	$N = 1000$
105	28.9	27.7	25.3	46.2
108	44.9	31.9	37.0	46.2
110	46.2	46.2	46.2	46.1

pump 1's maximum flow rate is above this line there are feasible solutions; however, they become more limited as consumer power demand increases. For example, when the consumer power demand level is 115%, pump 1's maximum flow rate (red) is below the gray line for all pump 2 flow rates and so there is no feasible solution.

C. Control Policy Parameters

We next solved the chance-constrained optimization problem for each case and consumer power demand level. The time to solve the problem ranged from less than a second when $N = 10$ to around a minute when $N = 1000$. We report average control policy parameters in Table II. As the consumer power demand increases the set of feasible operating points reduces (as seen in Fig. 3), forcing pump 1 to point C. However, at point C pump 1 has more flexibility to respond to water demand forecast error than when it is at point A. Therefore, its control policy parameter increases to 46%. For $N = 1000$ we find that, usually, at least one scenario requires pump 1 to operate at point C and so pump 1's control policy parameter is approximately 46%. We see more variation across solution instances as N and consumer power demand decreases. For $N < 1000$, pump 1's average control policy parameter increases with increasing power demand. We do not observe a clear upwards or downwards trend in pump 1's average control policy parameter as N increases.

D. Feasibility Range

In order to evaluate the solutions obtained for each case and consumer power demand level, we tested them using

TABLE III
VIOLATION PROBABILITY (%), MEAN μ AND STANDARD DEVIATION σ

Power Demand (%)	$N = 10$		$N = 50$		$N = 100$		$N = 1000$	
	μ	σ	μ	σ	μ	σ	μ	σ
105	8.40	9.83	3.85	3.81	1.86	1.06	1.32	0.13
108	2.27	6.49	1.47	1.19	1.19	0.36	1.11	0.07
110	1.52	3.55	1.17	0.11	1.20	0.12	1.14	0.13

10,000 new sets of randomly generated water demands for all junctions. Specifically, for each solution instance and set of randomly generated water demands we determined if the pumping schedule and real-time pump adjustments computed from the control policy parameters would lead to water and power flows that satisfy the WDN and PDN constraints. Table III shows the average and the sample standard deviation of the violation probabilities. As expected, the empirical violation probability generally decreases as N increases. We observe a slight increase in violation probability despite an increase in scenarios from $N = 50$ to 100 for the 110% consumer power demand level; however, this is due the specific scenarios drawn in those cases. Averaging more solution instances would likely eliminate this issue.

We observe that the heuristically-applied scenario approach is effective solving our problem and finding schedules and control policy parameters that lead to feasible solutions despite water demand forecast error. Empirically, we find that only $N = 100$ scenarios are needed to consistently achieve violation probabilities less than 5% for all consumer power demand levels. This is significantly less than the $N = 4,621$ scenarios computed with (17). Similarly, only $N = 1000$ scenarios are needed to consistently achieve violation probabilities less than 2%, as compared to $N = 11,552$ scenarios computed with (17). We also find that the violation probability varies with the consumer power demand level, where solutions computed for higher power demands generally have lower violation probabilities. We note that our results are not generalizable to other coupled power and water distribution networks; however, this preliminary investigation shows that our approach is worth testing on realistic networks. We are currently exploring the scalability of this approach to larger networks.

V. CONCLUSION

In this paper, we explored the performance of a heuristic scenario-based approach on a nonconvex chance-constrained optimal water pumping problem including water and power network constraints and water demand forecast error. By including power distribution network constraints in the problem, we can ensure that water pumping does not exacerbate voltage issues in power distribution networks. We computed both the optimal pumping schedule and the parameters of an affine control policy that would be used to modify pumping to respond to water demand forecast errors in real-time. We found that the approach can achieve low chance constraint violation probabilities with a relatively small number of scenarios. However, we noted several challenges, namely that finding a solution requires reinitializing the nonlinear solver multiple times and no guarantee that the solution is the global

minimum. Moreover, the approach confers no probabilistic guarantees when applied to a nonconvex problem. In future work, we will explore alternative solution approaches such as scenario-based methods tailored to nonconvex problems.

REFERENCES

- [1] International Energy Agency, *World Energy Outlook 2016*, Paris, 2016.
- [2] I. Vijay, A. Aghajanzadeh, and O. Schetrit, "Low-Cost, Scalable, Fast Demand Response for Municipal Wastewater and Recycling Facilities," California Energy Commission, Tech. Rep. CEC-500-2015-086, 2015.
- [3] P. W. Jowitt and G. Germanopoulos, "Optimal pump scheduling in water-supply networks," *J. Water Resour. Plann. Manage.*, vol. 118, no. 4, pp. 406–422, 1992.
- [4] D. Cohen, U. Shamir, and G. Sinai, "Optimal operation of multi-quality water supply systems-II: the Q-H model," *Eng. Opt.*, vol. 32, no. 6, pp. 687–719, 2000.
- [5] R. Salgado, E. Todini, and P. O'Connell, *Comparison of the Gradient Method with some Traditional Methods for the Analysis of Water Supply Distribution Networks*. Taunton, UK: Research Studies Press Ltd., 1987, pp. 38–62.
- [6] H. Mala-Jetmarova, N. Sultanova, and D. Savic, "Lost in optimisation of water distribution systems? A literature review of system operation," *Environmental Modelling and Software*, vol. 93, pp. 209–254, 2017.
- [7] A. Babayan, Z. Kapelan, D. Savic, and G. Walters, "Least-Cost Design of Water Distribution Networks under Demand Uncertainty," *J. Water Resour. Plann. Manage.*, vol. 131, no. 5, pp. 375–382, 2005.
- [8] A. P. Goryashko and A. S. Nemirovski, "Robust energy cost optimization of water distribution system with uncertain demand," *Autom. Remote Control*, vol. 75, no. 10, pp. 1754–1769, 2014.
- [9] D. K. Molzahn and I. A. Hiskens, "A survey of relaxations and approximations of the power flow equations," Invited submission to *Foundations and Trends in Electric Energy Systems*, 2017.
- [10] E. Dall'Anese, K. Baker, and T. Summers, "Chance-Constrained AC Optimal Power Flow for Distribution Systems With Renewables," *IEEE Trans. Power Syst.*, vol. 32, no. 5, pp. 3427–3438, 2017.
- [11] R. Menke, E. Abraham, P. Pappas, and I. Stoianov, "Demonstrating demand response from water distribution system through pump scheduling," *Applied Energy*, vol. 170, pp. 377–387, 2016.
- [12] K. Oikonomou, M. Parvania, and R. Khatami, "Optimal Demand Response Scheduling for Water Distribution Systems," *IEEE Trans. on Industrial Electronics*, 2018.
- [13] K. Oikonomou and M. Parvania, "Optimal Coordination of Water Distribution Energy Flexibility with Power Systems Operation," *IEEE Trans. Smart Grid*, 2018.
- [14] A. Zamzam, E. Dall'Anese, C. Zhao, J. Taylor, and N. Sidiropoulos, "Optimal Water-Power Flow Problem: Formulation and Distributed Optimal Solution," *IEEE Trans. Control Netw. Syst.*, pp. 1–16, 2017.
- [15] D. Fooladivanda and J. A. Taylor, "Energy-optimal pump scheduling and water flow," *IEEE Trans. Control Netw. Syst.*, vol. 5, no. 3, pp. 1016–1026, Sept 2018.
- [16] Q. Li, S. Yu, A. Al-Sumaiti, and K. Turitsyn, "Modeling and co-optimization of a micro water-energy nexus for smart communities," in *2018 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe)*, Oct 2018.
- [17] P. R. Bhav and R. Gupta, *Analysis of Water Distribution Networks*. Oxford, U.K.: Alpha Science International Ltd., 2006.
- [18] M. E. Baran and F. F. Wu, "Optimal capacitor placement on radial distribution systems," *IEEE Trans. Power Del.*, vol. 4, no. 1, pp. 725–734, Jan 1989.
- [19] M. C. Campi, S. Garatti, and M. Prandini, "The scenario approach for systems and control design," *Annual Reviews in Control*, vol. 33, no. 2, pp. 149–157, 2009.
- [20] S. Grammatico, X. Zhang, K. Margellos, P. Goulart, and J. Lygeros, "A scenario approach for non-convex control design," *IEEE Trans. Autom. Control*, vol. 61, no. 2, pp. 334–345, Feb 2016.
- [21] W. H. Kersting, "Radial distribution test feeders," in *IEEE PowerEng. Soc. Winter Meet.*, vol. 2, 2001, pp. 908–912.
- [22] J. Burkardt, "The Truncated Normal Distribution," pp. 1–35, 2014. [Online]. Available: <http://people.sc.fsu.edu/~jburkardt/presentations/truncated>
- [23] A. Wächter and L. T. Biegler, "On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming," *Mathematical Programming*, vol. 106, no. 1, pp. 25–57, 2006.